Wishlist

Wish it. Snap it. Chase it. Grant it.

Software Requirements Document

Cross-platform mobile application (Flutter) with Python API backend

Document	SRD-WISH-001
Version	1.0 (Release for development)
Date	5 November 2026
Status	Approved for build
Audience	Flutter frontend team, Python backend team, QA, DevOps, design
Primary market	Ghana (Android first), designed to scale globally

Document Control

This document is the single source of truth for what the Wishlist system must do and how it must behave. Where this document and any verbal instruction conflict, this document wins until it is formally revised. Every requirement carries a unique identifier so that developers, testers and the product owner can refer to it unambiguously in tickets, pull requests and test cases.

Version	Date	Author	Description
0.1	1 Nov 2026	Product owner	Initial concept and feature list
0.5	3 Nov 2026	Product owner	Motion, typography and gamification sections added
1.0	5 Nov 2026	Product owner	Full SRD released to frontend and backend teams

How to read this document

Term	Meaning
shall	Mandatory. The system will not be accepted without it.
should	Strongly recommended. Omit only with written approval from the product owner.
may	Optional. Implement if time allows.
M / S / C	Priority: Must have (MVP), Should have (Release 1.1), Could have (Release 1.2 and later).
FR / NFR / UI / API	Functional requirement, non-functional requirement, UI and motion requirement, API requirement.

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1. Introduction

1.1 Purpose

This Software Requirements Document (SRD) defines the complete functional, non-functional, interface, data, motion-design and operational requirements for **Wishlist**, a mobile application in which people collect the things they wish for as pictures, arrange them on playful boards, save toward them, share them with people they trust and celebrate when a wish comes true. The document is written so that a Flutter developer and a Python developer can each start building their half of the system from it without needing further clarification.

1.2 Product vision

Most wish list tools feel like spreadsheets: grey rows, tiny thumbnails, no emotion. Wishlist turns wanting something into a small daily joy. Every tap should feel physical, every screen should move, and every milestone should feel like a celebration. The product must be memorable enough that people open it just to play with it, and useful enough that they come back to actually get the things they wished for.

1.3 Scope

In scope for Release 1.0 (MVP):

- Account creation and sign-in (phone OTP, email, Google, Apple).
- Multiple wish lists, each containing image-based wishes.
- Image capture, gallery import, crop, sticker-style cut-out and frame styles.
- List view and the free-form Vision Board (collage) view.
- Wish Jar (shake to pick a random wish).
- Granted flow with full-screen celebration, and the Granted Shelf.
- Savings tracker with the Reveal Meter (image sharpens as savings grow).
- Private link sharing and secret gift claiming.
- Streaks, Stardust (XP), levels and badges.
- Offline-first operation with background sync.
- Push and local notifications.
- Themes, font packs, sound and haptic settings, accessibility modes.
- Admin console for moderation and analytics.

Out of scope for Release 1.0: in-app purchases of physical goods, mobile money contributions from friends (planned for 1.2), web client, price scraping from shop URLs (planned for 1.1), augmented reality preview, and multi-language packs beyond English (planned for 1.1).

1.4 Definitions and abbreviations

Term	Definition
Wish	A single desired item, represented by one or more images plus optional details.
List	A named collection of wishes (for example Birthday, Someday, Gifts for Mum).
Vision Board	Free-form collage view of a list where wishes can be dragged, scaled and rotated.
Wish Jar	Shake-to-reveal feature that surfaces one random open wish.
Granted	State of a wish that has come true. Triggers celebration.
Reveal Meter	Savings progress mechanic in which a wish image goes from blurred to sharp.
Stardust	In-app experience points earned for actions.
Claim	A secret mark placed by a friend to say they will buy a wish for the owner.
Owner	The user who created a list. Collaborator and Viewer are defined in section 4.
OTP	One-time password sent by SMS.
GHS	Ghana cedi, the default currency.
MoMo	Mobile money (MTN MoMo, Telecel Cash, AirtelTigo Money).
API	Application programming interface (REST over HTTPS, JSON).
SRD	Software Requirements Document (this document).

1.5 References

- Ghana Data Protection Act, 2012 (Act 843) and Data Protection Commission guidelines.
- Flutter documentation, Material 3 and platform channel guidelines.
- OWASP Mobile Application Security Verification Standard (MASVS) and OWASP API Security Top 10.
- WCAG 2.2 level AA (adapted for mobile).
- Google Play Developer Program Policies and Apple App Store Review Guidelines.
- RFC 6749 (OAuth 2.0), RFC 7519 (JWT), RFC 9110 (HTTP semantics).

2. Overall Description

2.1 Product perspective

Wishlist is a new, standalone product. It consists of a Flutter mobile client for Android and iOS, a Python REST API backend with a relational database, an object store for images, a background worker for image processing and notifications, and a small web-based admin console. It integrates with third-party services for SMS OTP, push notifications, social sign-in, error monitoring and analytics.

2.2 Target users and personas

Persona	Description	What they need
Ama, 21, university student, Accra	Android phone, mid-range, uses mobile data bundles carefully. Saves for a laptop, sneakers and a trip.	Light data use, offline use, a fun board she can screenshot and post, a way to save toward a goal.
Kwame, 29, young professional, Kumasi	Plans gifts and personal buys. Shares lists with his partner and family.	Private sharing, secret claims so he never gets duplicate gifts, price notes in GHS.
Efua, 35, mother of two, Takoradi	Keeps a list for each child and a family list. Uses WhatsApp heavily.	Easy sharing over WhatsApp, simple lists, child-safe visuals.
Daniel, 24, diaspora (London)	Wants to send gifts home and see what relatives wish for.	Shared lists across countries, multi-currency display, secure claims.

2.3 Operating environment

Item	Requirement
Android	Android 8.0 (API 26) and above. Primary target. Must run smoothly on devices with 3 GB RAM.
iOS	iOS 14 and above. iPhone 8 and newer.
Network	2G/3G/4G/5G and Wi-Fi. The app must remain usable offline.
Backend hosting	Linux containers (Docker) on a cloud provider with a nearby region (for example Europe/Cape Town) and CDN edge nodes serving West Africa.
Admin console	Latest two versions of Chrome, Edge, Firefox and Safari on desktop.

2.4 Design and implementation constraints

- Frontend shall be written in Flutter (stable channel, Dart 3.x) as a single code base.
- Backend shall be written in Python 3.12 or later using FastAPI.

- All client-to-server traffic shall use HTTPS with TLS 1.2 or higher.
- The app must stay within the performance budgets in section 11 while delivering the rich animation described in section 6.
- Image storage shall use an S3-compatible object store so the provider can change without code change.

2.5 Assumptions and dependencies

- Users have a smartphone with a camera and a working SMS number or email address.
- An SMS gateway that delivers reliably to Ghanaian networks (MTN, Telecel, AirtelTigo) is available.
- Firebase Cloud Messaging (FCM) and Apple Push Notification service (APNs) remain available.
- Fonts, icon packs and animation files used in the app are licensed for commercial use.
- App Store and Google Play accounts are provisioned by the product owner.

3. System Architecture

3.1 Architecture overview

Wishlist follows a client-server architecture. The Flutter client is offline-first: it reads and writes to a local database and synchronises with the API in the background. The API is stateless and horizontally scalable. Heavy work (image processing, notification fan-out, badge evaluation) runs in background workers so API responses stay fast.

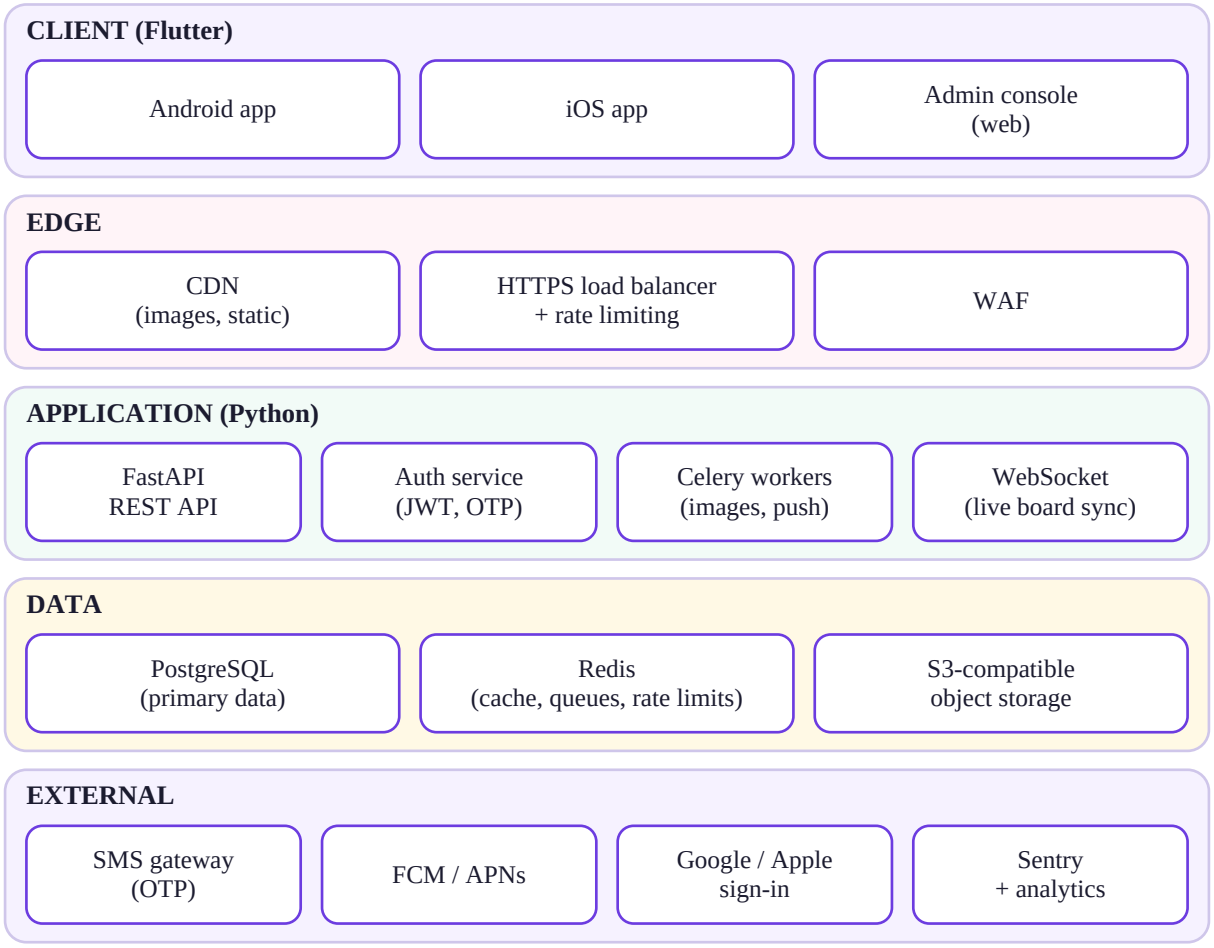


Figure 1. Logical architecture of the Wishlist system.

3.2 Technology stack

Layer	Technology	Notes
Mobile client	Flutter 3.x, Dart 3.x	Single code base for Android and iOS.
State management	Riverpod (with code generation)	Predictable, testable, no BuildContext dependency.
Navigation	go_router	Deep links for shared lists and notifications.
Local database	Drift (SQLite) with SQLCipher option	Offline-first storage and sync queue.
Networking	dio with interceptors	Auth refresh, retry with back-off, logging.
Animation	flutter_animate, Rive, Lottie, custom painters, Flutter physics (spring simulations)	See section 6.
Backend framework	FastAPI (Python 3.12), Pydantic v2	Async, auto-generated OpenAPI documentation.
ORM and migrations	SQLAlchemy 2.x, Alembic	Versioned schema migrations.
Database	PostgreSQL 15 or later	Primary relational store.
Cache and queue	Redis 7	Cache, rate limiting, Celery broker.
Workers	Celery	Image pipeline, notifications, badges, clean-up jobs.
Object storage	S3-compatible (AWS S3, Cloudflare R2 or MinIO)	Pre-signed uploads, CDN in front.
Image processing	Pillow, pillow-heif, optional rembg (ONNX) worker	Resize, WebP conversion, cut-out stickers, blur hash.
Auth	JWT access and refresh tokens, Argon2id password hashing	OAuth for Google and Apple.
Push	Firebase Cloud Messaging	Android and iOS via FCM and APNs.
Observability	Sentry, Prometheus, Grafana, structured JSON logs	Alerts to the on-call developer.
CI/CD	GitHub Actions, Docker, Fastlane	Automated tests, builds and store uploads.

3.3 Deployment environments

Environment	Purpose	Data
Local	Developer machines using Docker Compose.	Seed and fake data only.
Development	Continuous integration target; merged work lands here automatically.	Synthetic data.
Staging	Mirror of production for QA, UAT and store review builds.	Anonymised data only.
Production	Live users.	Real data, encrypted backups.

4. User Roles and Permissions

Roles are assigned per list for end users, and globally for staff. A user may be Owner of some lists and Viewer of others.

Role	Scope	Capabilities
Guest	Global	Can browse onboarding and open a public shared list link in read-only mode. Cannot create data until they sign up.
Owner	Per list	Full control: edit, delete, change privacy, invite, remove people, grant wishes, see savings. Cannot see who claimed a wish.
Collaborator	Per list	Can add, edit and re-arrange wishes on the list. Cannot delete the list or change privacy.
Viewer	Per list	Can view wishes and react with emoji. May claim a wish secretly (hidden from the Owner).
Moderator	Global (staff)	Reviews reports, hides or removes content, suspends accounts temporarily.
Admin	Global (staff)	All Moderator rights plus user management, feature flags, badge catalogue, analytics, audit log.

5. Functional Requirements

Requirements are grouped into modules. Each module states its purpose, then lists numbered requirements with priority. Section 6 adds the motion and visual requirements that apply to every module.

5.1 Module A: Onboarding and Authentication

Purpose: get a new user from install to their first wish in under 60 seconds.

ID	Requirement (the system shall...)	Pri.
FR-A1	show a 3-screen animated onboarding (what Wishlist is, how to add a wish, how to celebrate) that can be skipped at any time and is shown only on first launch.	M
FR-A2	allow sign-up and sign-in with a Ghana or international phone number (E.164 format) using an SMS one-time password valid for 5 minutes, with a maximum of 5 attempts per code.	M
FR-A3	allow sign-up and sign-in with email and password. Passwords shall be at least 8 characters and checked against a list of common passwords.	M
FR-A4	allow sign-in with Google (Android and iOS) and Sign in with Apple (iOS).	M
FR-A5	allow the user to start in guest mode, create up to 5 wishes locally, and later convert to a full account without losing data.	S
FR-A6	let the user choose a display name, an avatar (photo or one of 24 animated characters) and a favourite colour during profile setup.	M
FR-A7	support password reset by email link or by SMS OTP.	M
FR-A8	support optional biometric unlock (fingerprint or face) to open the app, using the platform biometric API.	S
FR-A9	allow users to link or unlink additional sign-in methods from Settings, provided at least one method remains.	S
FR-A10	allow account deletion from Settings. Deletion shall remove all personal data within 30 days and confirm by notification.	M
FR-A11	allow users to export all their data (lists, wishes, images) as a ZIP file containing JSON and original images.	S

5.2 Module B: Lists

ID	Requirement (the system shall...)	Pri.
FR-B1	let a user create unlimited lists (soft limit of 50 per account, adjustable by feature flag), each with a name (1 to 40 characters), an emoji or icon, a cover colour or cover image, and an optional description.	M
FR-B2	provide three default list templates on first run: Someday, Birthday and Gifts for Others, which the user may rename or delete.	S
FR-B3	let the user re-order lists by long-press and drag, with a physical lift, tilt and drop animation.	M
FR-B4	let the user archive, restore, duplicate and delete lists. Deleting a list shall move it to a Bin for 30 days before permanent deletion.	M
FR-B5	show on each list card the number of wishes, the number granted, and a mini preview collage of up to 4 wish images.	M
FR-B6	support list privacy levels: Private (only owner), Invite only (named people), and Link (anyone with the secret link).	M
FR-B7	let the user pin up to 3 lists to the top of the home screen.	S
FR-B8	let the user set an optional target date on a list (for example a birthday) and show a live countdown with animated digits.	S

5.3 Module C: Wishes

A wish is the heart of the product. Each wish must feel like a collectible card.

ID	Requirement (the system shall...)	Pri.
FR-C1	let the user create a wish by (a) taking a photo, (b) choosing one or more images from the gallery, (c) pasting an image from the clipboard, or (d) receiving an image from another app via the system Share sheet.	M
FR-C2	allow up to 5 images per wish, with the first image used as the cover. Images may be re-ordered by drag.	M
FR-C3	store for each wish: title (optional, 0 to 80 characters), note (0 to 1000 characters), price (optional, decimal), currency (default GHS), priority (1 to 5 stars), category, tags (up to 10), product link (optional URL), and a free-text field called Why I want this.	M
FR-C4	provide a quick-add mode in which the user can add a wish with an image only. The title shall default to an empty value and the user may fill in details later.	M
FR-C5	offer an image editor on add: crop (free, square, 4:5, 9:16), rotate, flip, brightness, and 12 filters.	M
FR-C6	offer a Sticker Cut-out tool that removes the image background (on-device when supported, otherwise via the backend worker) and lets the wish appear as a die-cut sticker with a white outline.	S

ID	Requirement (the system shall...)	Pri.
FR-C7	offer 10 frame styles for a wish card (polaroid, torn paper, neon, gold, film strip, comic, postage stamp, sticky note, ticket, plain). Frames shall be applied at display time and never baked into the stored image.	M
FR-C8	let the user choose a mood sticker for each wish (for example fire, dream, need, treat, someday) shown as an animated badge on the card.	S
FR-C9	support edit, duplicate, move to another list, copy to another list, and delete for a wish. Deleting shall be undoable for 10 seconds through a snackbar and recoverable from the Bin for 30 days.	M
FR-C10	support multi-select (long-press) for bulk move, bulk delete, bulk tag and bulk grant.	S
FR-C11	support voice notes of up to 30 seconds attached to a wish, stored as compressed audio.	C
FR-C12	automatically detect a link in the note or clipboard and, when connected, offer to fetch the product title, price and primary image from the page (Release 1.1).	C
FR-C13	reject images that are corrupted, are larger than 25 MB before compression, or are not JPEG, PNG, WebP or HEIC, with a friendly animated error.	M

5.4 Module D: Views, Search and Filters

ID	Requirement (the system shall...)	Pri.
FR-D1	provide four views for each list, switchable with an animated morphing toggle: Grid (two-column staggered cards), Stack (swipeable deck), Vision Board (free collage) and Timeline (by date added or target date).	M
FR-D2	in Vision Board view allow the user to drag, pinch-scale, rotate, bring forward and send backward each wish, and to place free text, emoji stickers and washi tape. Positions, scale, rotation and z-order shall be saved per list.	M
FR-D3	support a Vision Board canvas that is larger than the screen and can be panned and zoomed between 0.5x and 3x with inertia.	S
FR-D4	provide an Auto-arrange button that tidies the board with a springy animation into a collage or grid layout.	S
FR-D5	allow the user to export a Vision Board as a PNG or a 9:16 story image and share it through the system Share sheet. Export resolution shall be at least 1080 by 1920 pixels.	M
FR-D6	provide global search across lists, wish titles, notes and tags with results appearing as the user types (debounced to 250 ms) and highlighting of matches.	M
FR-D7	provide filters (open, granted, priority, category, price range, has image count, claimed by someone) and sorts (newest, oldest, priority, price low to high, price high to low, nearest to goal).	M
FR-D8	provide a Stack view in which swiping right marks a wish as Still want it, swiping left sends it to Maybe Later, and swiping up starts the Grant flow.	S

5.5 Module E: Wish Jar (Shake)

ID	Requirement (the system shall...)	Pri.
FR-E1	show a Wish Jar tab with a glass jar animation containing floating mini wish images.	M
FR-E2	when the device is shaken (accelerometer threshold above 2.7 g sustained for 400 ms) or the jar is tapped, play a shake animation with haptic feedback and then pop one random open wish out of the jar with a spring animation and sound.	M
FR-E3	let the user choose which lists feed the jar, and weight selection by priority (5-star wishes are more likely).	S
FR-E4	offer three actions on the revealed wish: Do something about it (opens the wish), Put it back, and Grant it.	M
FR-E5	provide a Daily Wish: once per day at a user-selected time the app sends a notification with a mystery wish that is revealed when opened.	S

5.6 Module F: Granting and Celebration

ID	Requirement (the system shall...)	Pri.
FR-F1	let the user mark a wish as Granted from the wish screen, the Stack view or the Jar, optionally adding the date, the price actually paid, a photo of the real item and a short memory note.	M
FR-F2	on grant, play a full-screen celebration of at least 2.5 seconds: confetti cannon, a stamp that slams over the image reading GRANTED, screen shake, haptic pattern and a sound effect, followed by the card flying to the Granted Shelf.	M
FR-F3	provide a Granted Shelf: a 3D-style trophy shelf displaying granted wishes as glossy cards, with total number granted and total value (if prices were given).	M
FR-F4	allow a granted wish to be un-granted within 24 hours.	S
FR-F5	generate a shareable Granted card image (before and after side by side) for sharing on social media.	S
FR-F6	provide a yearly Wrapped summary in December showing wishes added, granted, money saved, longest streak and top category, as an animated story with swipe navigation.	C

5.7 Module G: Savings and the Reveal Meter

Wishlist does not hold money. The savings feature is a personal tracker that the user updates manually. It is intended to motivate saving, not to process payments.

ID	Requirement (the system shall...)	Pri.
FR-G1	let the user set a savings goal for any wish that has a price. The goal defaults to the wish price.	M

ID	Requirement (the system shall...)	Pri.
FR-G2	let the user add and remove savings entries (amount, date, optional note) and show the running total and percentage.	M
FR-G3	display the wish cover image under a blur and greyscale effect that reduces continuously with progress. At 0 percent the image is fully blurred, at 100 percent it is sharp and in colour. The change shall animate whenever an entry is added.	M
FR-G4	show milestone pop-ups at 25, 50, 75 and 100 percent with a short celebration, and automatically offer the Grant flow at 100 percent.	M
FR-G5	support Save Streak reminders: the user may set a weekly or monthly target amount and receive reminders.	S
FR-G6	show an estimated completion date based on the average saving rate of the last 30 days.	S
FR-G7	support currencies GHS (default), USD, GBP, EUR, NGN and a generic custom currency. Display only; no conversion is required for Release 1.0, though exchange rates may be added in 1.1.	M
FR-G8	provide an optional Round-up Jar mode where the user can log small amounts toward any wish from a single quick-add button.	C

5.8 Module H: Sharing, Collaboration and Secret Claims

ID	Requirement (the system shall...)	Pri.
FR-H1	generate a unique, unguessable share link (at least 128 bits of entropy) for each list on request. The Owner may revoke or regenerate the link at any time, instantly invalidating the old one.	M
FR-H2	open shared links in the app if installed, otherwise show a lightweight web preview with a Get the app button (deep link / universal link / Android App Link).	M
FR-H3	let the Owner invite people by phone number, email or link, and assign the Collaborator or Viewer role.	M
FR-H4	let a Viewer claim a wish with a single tap (I will get this). The claim shall be hidden from the Owner. Other viewers shall see the wish as Claimed without seeing who claimed it, unless the claimer chooses to reveal their name to other viewers.	M
FR-H5	let the claimer release their claim at any time, and automatically release claims after the Owner marks the wish as Granted.	M
FR-H6	allow Viewers and Collaborators to react to a wish with one of 8 emoji reactions and short comments (up to 200 characters). Comments shall be visible to the Owner unless sent as a Secret Note to other viewers only.	S
FR-H7	provide real-time collaborative editing on shared Vision Boards, showing other collaborators as coloured cursors with name tags, with last-write-wins conflict handling per object.	S
FR-H8	let users report a list, wish or comment for abuse. Reports shall reach the moderation queue within 5 seconds.	M
FR-H9	let users block another user, which hides their content and prevents invitations and comments.	M
FR-H10	allow friends to chip in toward a wish using mobile money (MTN MoMo, Telecel Cash, AirtelTigo Money) through a licensed payment aggregator (Release 1.2).	C

5.9 Module I: Gamification

Gamification must reward healthy engagement (adding wishes, saving, granting) and must never pressure users into spending money.

ID	Requirement (the system shall...)	Pri.
FR-I1	award Stardust for actions according to the table below, capped at 500 Stardust per day to prevent farming.	M
FR-I2	track a Wish Streak: consecutive days on which the user performs at least one qualifying action. Provide one free Streak Freeze per month and a visible flame animation that grows with the streak.	M

ID	Requirement (the system shall...)	Pri.
FR-I3	provide 30 levels with named titles (for example Dreamer, Collector, Dream Chaser, Wish Wizard) and a level-up animation with a new frame or sticker unlocked at certain levels.	S
FR-I4	provide at least 40 badges (catalogue in Appendix C) with locked silhouettes, unlock animations and a Trophy Room screen.	S
FR-I5	show a weekly Wish Challenge (for example add 3 wishes with photos, save toward a wish twice) with a progress ring.	C
FR-I6	allow users to turn off all gamification elements in Settings for a calm mode.	M

Action	Stardust	Limit
Add a wish with an image	10	No limit within the daily cap
Fill in title, price and note for a wish	5	Once per wish
Log a savings entry	15	3 per day
Reach a savings milestone	50	Per milestone
Grant a wish	100	No limit
Share a list	20	Once per list per day
Daily open (streak)	5 + streak bonus	Once per day
Use the Wish Jar	5	3 per day

5.10 Module J: Notifications

ID	Requirement (the system shall...)	Pri.
FR-J1	support push notifications (FCM) and local scheduled notifications. The user shall be asked for notification permission with a friendly pre-permission screen at a meaningful moment, not at first launch.	M
FR-J2	send notifications for: Daily Wish, streak at risk (only if the user opted in), savings reminders, list invitation, a new comment or reaction, a target-date countdown, and a milestone achieved.	M
FR-J3	never send a notification that reveals who claimed a wish or that a claim exists to the Owner.	M
FR-J4	provide per-category toggles and quiet hours (default 22:00 to 07:00).	M
FR-J5	limit marketing and engagement notifications to a maximum of one per day, and respect operating-system do-not-disturb.	M
FR-J6	provide an in-app Inbox screen retaining notifications for 30 days.	S

5.11 Module K: Settings, Themes and Accessibility

ID	Requirement (the system shall...)	Pri.
FR-K1	offer at least 8 colour themes including light, dark and AMOLED black, plus 6 playful seasonal themes. The theme shall switch with a circular reveal animation from the toggle position.	M
FR-K2	offer at least 6 font packs the user can choose in Settings (for example Bubbly, Handwritten, Retro Arcade, Elegant, Comic, Clean). Fonts shall be bundled or downloaded once and cached.	M
FR-K3	offer an icon pack choice (Rounded, Sticker, Neon) and an optional alternate app icon (at least 5 variants).	S
FR-K4	offer separate toggles for sound effects, haptics, animations level (Full, Reduced, Off) and confetti.	M
FR-K5	respect the operating-system Reduce Motion and Bold Text settings, automatically switching to the Reduced animation level.	M
FR-K6	support dynamic type up to 200 percent without layout breaking, and screen readers (TalkBack and VoiceOver) with meaningful labels for every interactive element and image.	M
FR-K7	provide a Data Saver mode that lowers image quality, disables auto-play animations that use large assets, and syncs only on Wi-Fi.	M
FR-K8	provide Help, Send feedback (with optional screenshot), Terms of Service, Privacy Policy and Licences screens.	M

5.12 Module L: Offline Mode and Synchronisation

ID	Requirement (the system shall...)	Pri.
FR-L1	allow the user to read, create, edit and delete lists and wishes while offline. Changes shall be queued locally and synchronised automatically when connectivity returns.	M
FR-L2	queue image uploads and retry with exponential back-off (2, 4, 8, 16, 32 seconds up to 5 minutes) with resume capability for interrupted uploads.	M
FR-L3	show a subtle sync indicator (animated cloud) with states: synced, syncing, offline, error. The user shall never be blocked from using the app because of sync.	M
FR-L4	resolve conflicts using per-field last-write-wins with server-assigned timestamps, except for Vision Board objects which use per-object last-write-wins and for savings entries which are append-only.	M
FR-L5	cache viewed images in a disk cache (default 300 MB, user-adjustable from 50 MB to 1 GB) with least-recently-used eviction.	M
FR-L6	support multi-device use: changes made on one device shall appear on another within 10 seconds when both are online.	S

5.13 Module M: Admin Console and Moderation

ID	Requirement (the system shall...)	Pri.
FR-M1	provide a web admin console with role-based login, two-factor authentication and an audit log of all staff actions.	M
FR-M2	provide a moderation queue listing reported items with preview, reporter reason, history and actions (dismiss, hide, remove, warn user, suspend user).	M
FR-M3	automatically screen uploaded images with an automated moderation service for nudity, violence and illegal content, and quarantine flagged images pending review.	M
FR-M4	provide user search and management (view profile metadata, suspend, restore, trigger data export or deletion) without exposing private list content unless a report requires it.	M
FR-M5	provide a dashboard with daily active users, new sign-ups, wishes created, wishes granted, retention (D1, D7, D30), crash-free rate and storage used.	S
FR-M6	provide feature flags and remote config (for example enable Wrapped, change Stardust values, change soft limits) without an app release.	S
FR-M7	provide a badge, theme and template catalogue editor.	C

6. UI, Visual Design and Motion Requirements

This section is the heart of the "fun" promise. The product owner's brief is explicit: the app must feel wild, bouncy and alive, with crazy animations, expressive fonts, delightful transitions and a unique icon language. The requirements below turn that brief into measurable rules the frontend team can build and the QA team can test.

6.1 Design principles

Principle	What it means in practice
Everything is physical	Elements have weight. They squash, stretch, bounce, lift off the page and cast shadows. Nothing just appears; it arrives.
Reward every tap	Every tappable element responds within 100 ms with a visual change, and usually a haptic tick.
Crazy, but never in the way	Animations are skippable, never block input for more than 400 ms (except the Granted celebration, which may be tapped to skip), and never hide information.
Wishes are the stars	Images are always the largest element on a screen. UI chrome stays quiet so the pictures can shine.
Consistent chaos	A shared motion system (section 6.5) ensures animations feel like one family, not random effects.
Inclusive by default	Reduced-motion, screen readers, large text and low-end devices are first-class, not afterthoughts.

6.2 Visual identity

The visual style is called **Sticker Pop**: thick rounded outlines, saturated colours, paper-cut shadows, tilted elements, and cards that look like stickers and polaroids stuck to a lively surface.

Token	Hex	Usage
Wish Violet	#6C3CE0	Primary brand colour, primary buttons, active tab.
Pop Pink	#E4357E	Accent, favourite and priority highlights, celebration stamp.
Sunny Gold	#F2B705	Stardust, streak flame, trophy elements, warnings.
Mint Green	#0E9F6E	Granted, success, savings progress.
Sky Blue	#4DB8FF	Info, links, collaborator cursors.
Midnight Ink	#1B1B2F	Text on light, dark theme background.
Paper Cream	#FFF9F0	Light theme background.
Cloud Lilac	#F6F2FF	Cards and subtle surfaces.

- Corner radius: 24 dp for cards, 16 dp for buttons and chips, fully round for avatars and the centre Add button.
- Outline: 2.5 dp solid Midnight Ink on stickers and cards in light theme. Elevation shall be expressed by an offset hard shadow (4 dp right, 4 dp down) rather than a soft blur on key elements.
- Spacing: 4 dp base grid. Standard screen padding 20 dp.
- Every theme shall pass a minimum text contrast ratio of 4.5:1 for body text and 3:1 for large text.

6.3 Typography and font packs

The app shall use expressive, characterful typefaces. All fonts listed are licensed under the SIL Open Font License and are free for commercial use. Fonts shall be bundled in the app assets (not fetched at runtime) for the default pack so the first screen never flashes unstyled text.

Role	Default font	Alternatives in font packs
Display (logo, big headings, celebration)	Bagel Fat One	Bangers (Comic), Press Start 2P (Retro Arcade), Playfair Display Black (Elegant)
Headings	Fredoka (SemiBold)	Baloo 2, Chewy, Lilita One
Body and UI	Nunito (Regular, SemiBold)	Quicksand, DM Sans (Clean pack)

Role	Default font	Alternatives in font packs
Handwritten notes and stickers	Caveat	Gochi Hand, Patrick Hand, Reenie Beanie
Numbers (prices, countdown, Stardust)	Fredoka	Rubik Mono One (Retro pack)

ID	Requirement (the system shall...)	Pri.
UI-T1	ship six font packs: Bubbly (default), Handwritten, Retro Arcade, Elegant, Comic and Clean. Switching pack shall update all text live without restarting the app.	M
UI-T2	use a type scale of 12, 14, 16, 20, 24, 32, 44 and 64 sp. Body text shall never be smaller than 14 sp.	M
UI-T3	animate display text on key moments with per-letter motion (letters bounce in with a 30 ms stagger, wobble on hover or press, and drop with gravity on exit).	M
UI-T4	support user-entered text in any Unicode script, including Twi and Ga characters (for example ɔ, ɛ, ŋ), with a fallback font that contains these glyphs.	M
UI-T5	render wish titles on cards with automatic fitting: shrink down to 14 sp and then truncate with an ellipsis; full text shall be available on the wish screen.	M

6.4 Iconography

ID	Requirement (the system shall...)	Pri.
UI-I1	use a custom icon set called Wishicons with at least 120 icons, drawn on a 24 by 24 grid with a 2 dp stroke and rounded joins. Three styles shall be provided: Rounded (outlined), Sticker (filled with outline and offset shadow) and Neon (glow).	M
UI-I2	deliver icons as an icon font and as SVG, and deliver the 20 most prominent icons also as Rive animations (see below).	M
UI-I3	animate navigation icons: when selected, each icon plays a unique one-shot animation (for example the jar lid pops, the star spins, the plus rotates 135 degrees into a cross, the trophy shines) and then rests in its active pose. Duration 400 to 600 ms.	M
UI-I4	animate action icons on tap: heart beats and releases tiny hearts, star bursts into sparkles, trash can lid flips open, share icon throws a paper plane, bell rings with swing and sound waves.	M
UI-I5	provide an animated empty-state illustration set (at least 12) in Lottie or Rive, each with a character called Wishy, a friendly floating star, in different moods (sleepy, excited, searching, celebrating, shy, confused).	M
UI-I6	provide 5 alternate app icons, delivered through the platform alternate-icon mechanism.	S

Required icon list (minimum): home, jar, add, shelf, profile, list, wish, star, heart, fire (streak), sparkle, camera, gallery, crop, filter, sticker, scissors, frame, trash, restore, edit,

duplicate, move, share, link, lock, unlock, gift, claim, comment, emoji, bell, calendar, tag, price, wallet, piggy bank, target, trophy, badge, level, crown, shake, dice, search, sort, filter, grid, stack, board, timeline, undo, redo, cloud (synced), cloud (syncing), cloud (offline), settings, theme, font, sound, vibration, accessibility, help, feedback, block, report, export, delete account.

6.5 Motion system

To keep craziness consistent, all animation shall be built from the shared motion tokens below. Developers shall implement them once in a central file (for example lib/core/motion/motion_tokens.dart) and shall not hard-code durations or curves in widgets.

Token	Value	Used for
dur.micro	120 ms	Tap feedback, colour changes, ripple.
dur.short	220 ms	Chips, toggles, small element moves.
dur.medium	380 ms	Card expand, list item insertion, tab change.
dur.long	600 ms	Page transitions, morphs, hero flights.
dur.hero	900 ms	Onboarding scenes, level-up, theme reveal.
dur.celebrate	2500 to 4000 ms	Granted celebration, badge unlock, milestones.
curve.pop	easeOutBack (overshoot 1.2)	Things appearing.
curve.rubber	elasticOut	Boing and wobble.
curve.smooth	easeInOutCubicEmphasized	Big moves and page slides.
curve.drop	bounceOut	Objects landing.
spring.soft	mass 1, stiffness 180, damping 12	Drag release, board objects, jar items.
spring.snappy	mass 1, stiffness 320, damping 20	Buttons, switches, tab pills.
stagger.list	40 ms per item (max 12 items, rest instant)	List and grid entrance.

ID	Requirement (the system shall...)	Pri.
UI-M1	implement motion tokens centrally and expose an animation level setting (Full, Reduced, Off) that scales durations (1.0, 0.5, 0) and disables decorative loops in Reduced and Off.	M
UI-M2	ensure every screen has an entrance animation, an idle life animation (for example floating stars, a breathing jar, a swaying flame) and an exit animation.	M
UI-M3	make every button squash on press (scale to 0.92 with spring.snappy) and rebound with overshoot on release, with a haptic light impact.	M

ID	Requirement (the system shall...)	Pri.
UI-M4	make every draggable object lift (scale 1.08, rotate 2 degrees, increase shadow) on pickup, follow the finger with slight lag and tilt according to velocity, and settle with spring.soft on release.	M
UI-M5	make all animations interruptible: starting a new gesture shall smoothly retarget the running animation without a jump.	M
UI-M6	ensure that no animation flashes more than 3 times per second or flashes large high-contrast areas (photosensitivity safety, WCAG 2.3.1).	M
UI-M7	pause or throttle idle animations when the app is in the background, the battery saver is on, the device is thermally throttled, or the screen is not the top route.	M

6.6 Transition catalogue

The following transitions are the official set. Each shall be implemented as a reusable PageRoute or widget so any screen can use it.

Name	Where used	Behaviour	Duration
Hero Fly	Wish card to Wish screen	The wish image flies, scales and straightens from the card into the detail header while the card's frame morphs into a full-bleed layout. Reverse on back.	600 ms
Page Curl Peel	Open a list	The list cover peels like a sticker from the home grid and the contents slide in underneath.	600 ms
Circular Reveal	Theme change, Granted celebration start	A circle expands from the touch point, painting the new theme or celebration backdrop.	900 ms
Jelly Slide	Tab change	The new page slides in with a jelly wobble (slight skew that settles). The active tab pill stretches like gum between tabs.	380 ms
Card Flip 3D	Wish front and back (details), Granted reveal	Perspective flip around the Y axis with a gloss highlight sweeping across.	500 ms
Pop-up Burst	Dialogs, bottom sheets	Sheets spring up with overshoot while the backdrop blurs. Dialogs pop from the tapped element.	380 ms
Confetti Wipe	After granting and onboarding finish	A band of confetti sweeps across the screen, and the next page is revealed behind it.	700 ms
Stack Shuffle	Switching to Stack view	Cards collapse into a deck then fan out with random small rotations.	700 ms
Morph Toggle	Switching view mode	The grid cards morph (position, size and rotation) to their positions in the new layout, without a fade.	600 ms

Name	Where used	Behaviour	Duration
Shake and Pop	Wish Jar reveal	Jar shakes violently, lid pops off, the chosen wish shoots out in an arc and lands with a bounce.	1800 ms
Stamp Slam	Granted	A huge GRANTED stamp scales from 3x to 1x with a heavy impact, screen shake and dust ring.	700 ms
Unblur Reveal	Savings progress	Blur radius and desaturation interpolate to the new value with a spring and a sparkle sweep.	900 ms
Fly to Shelf	After Granted	The card shrinks along a curved path to the Shelf tab icon, which bounces on arrival.	800 ms
Shared-axis Zoom	Settings, profile, secondary screens	Subtle zoom and fade for low-emphasis navigation to keep the app calm.	300 ms

6.7 Signature micro-interactions

Component	Interaction
Add (+) button	Idle: gently pulsing with a rotating sparkle ring. Press: squashes. Open: rotates 135 degrees while four option bubbles (Camera, Gallery, Paste, Quick note) fly out in an arc with spring stagger.
Wish card	Tilts toward the finger on press (parallax up to 8 degrees), with a gloss highlight moving across. Long-press: lifts and wobbles (iOS-style jiggle) to enter select or re-order mode.
Priority stars	Tapping a star fills all stars up to it with a sequential pop and tiny particle trails.
Pull to refresh	A star is pulled down on an elastic string and released into a firework when sync completes.
Empty list	Wishy floats around, yawns, and nudges the Add button with a speech bubble.
Streak flame	Size and colour grow with streak days (blue small flame to gold to rainbow at 100 days). Breaks with a puff of smoke if lost, and offers Streak Freeze.
Stardust counter	Numbers roll like a slot machine when Stardust increases; golden particles stream from the source of the reward to the counter.
Delete	The card crumples into a paper ball and is thrown into the bin with a bounce. Undo snackbar rewinds the animation.
Switches and checkboxes	Switch knob stretches in the direction of travel; checkbox draws its tick with a stroke animation and a small burst.
Loading	Never a plain spinner. Use a bouncing jar, spinning star or Wishy juggling images. Skeleton screens use a shimmer in brand colours.
Errors	Form fields shake horizontally; the error message drops in with a bounce; a sad Wishy sits on the page for fatal errors.
Typing in search	Letters appear with a typewriter pop; results rain down as stickers.
Scroll	Cards parallax at different speeds, the header collapses with a squash and the title letters re-flow.

6.8 Sound and haptics

Event	Sound (short, under 1.2 s)	Haptic
Tap button	Soft pop	Light impact
Add wish	Bubble plus sparkle	Medium impact
Delete	Paper crumple, thud	Heavy impact
Jar shake	Rattle of glass beads	Repeating light taps while shaking
Jar reveal	Cork pop and chime	Success pattern
Savings added	Coins drop	Light impact
Milestone	Ascending xylophone	Double medium impact
Granted	Fanfare and stamp thud, crowd cheer	Heavy impact then success pattern
Level up	Power-up arpeggio	Rising pattern
Error	Soft buzz	Error pattern

ID	Requirement (the system shall...)	Pri.
UI-S1	ship at least 25 original or licensed sound effects, each under 60 KB, compressed (AAC or OGG), mixed at a consistent loudness (about -16 LUFS) and played with a low-latency audio player.	M
UI-S2	never play sound when the device is on silent, and never override the user's music. Use ambient audio focus mode so background music continues at lowered volume only during celebrations if the user enabled it.	M
UI-S3	provide a haptics abstraction that maps semantic events (tap, success, warning, error, celebration) to platform patterns and is disabled when haptics are off or unsupported.	M

6.9 Animation technology and asset rules

Need	Technology	Rules
UI motion (fades, slides, scales, staggers)	flutter_animate and Flutter implicit and explicit animations	Use tokens. Prefer implicit animations for simple cases.
Physical movement (drag, jar items, board objects)	Flutter SpringSimulation and custom Ticker	Use spring.soft and spring.snappy.
Character animation and animated icons	Rive (state machines)	Each file under 150 KB. Inputs exposed to Dart for state changes.
Illustrations and one-shots	Lottie (dotLottie format preferred)	Each file under 200 KB. Pre-load on splash for the first screens.
Confetti, sparkles, particles	Custom painter particle system using a single canvas pass	Particle cap: 150 (high tier), 60 (mid), 20 (low). Pool and reuse objects.
Blur, ripple and glow effects	Fragment shaders (GLSL via Flutter FragmentProgram) where available	Fall back to ImageFiltered blur on unsupported devices.
3D-style tilts and flips	Transform with perspective matrix	No full 3D engine.

ID	Requirement (the system shall...)	Pri.
UI-A1	detect device performance class at first launch (RAM, CPU cores, Android performance class API, iOS model) and set the default animation tier: High, Medium or Low. The user may override it.	M
UI-A2	maintain 60 frames per second (and 90 or 120 on capable displays) during all navigation and drag interactions on a reference mid-range Android device (3 GB RAM). Jank (frames over 16.7 ms) shall be below 1 percent of frames during scripted benchmarks.	M
UI-A3	wrap independently animating regions in RepaintBoundary, avoid unnecessary saveLayer calls (opacity on large trees), and pre-warm shaders (use the Impeller renderer on iOS and on Android where stable).	M
UI-A4	dispose all animation controllers and Rive and Lottie resources when screens leave the tree. Memory shall not increase more than 15 MB across 50 repeated open-close cycles of any screen.	M
UI-A5	include a hidden debug overlay (performance, tier, frame times) available in debug and staging builds.	S

7. Screens and Navigation

7.1 Navigation structure

The app uses a bottom navigation bar with five items: **Home** (lists), **Jar**, a large central **Add** button, **Shelf** (granted wishes) and **Me** (profile, badges, settings). The bar itself is animated: the active pill stretches like gum, icons play their one-shot animation, and the bar hides on scroll down and returns on scroll up.

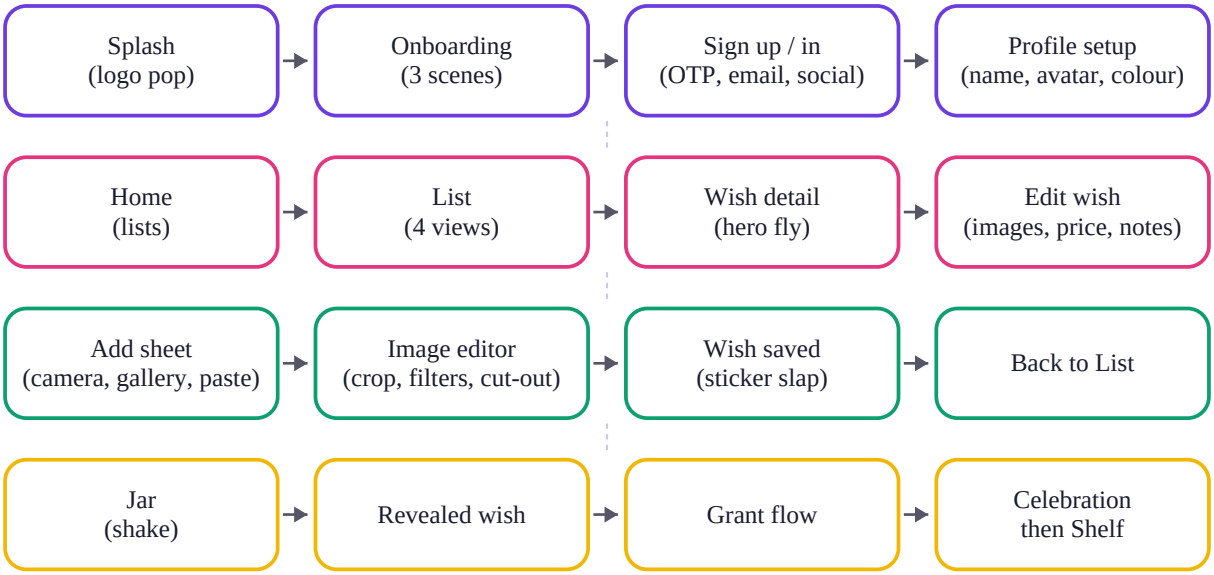


Figure 2. Primary navigation and key user flows.

7.2 Screen inventory

ID	Screen	Purpose and key elements	Signature motion
S01	Splash	Logo, loading of fonts and assets, auth check.	Letters of the logo drop in and bounce; a star shoots over.
S02	Onboarding	3 scenes with Skip and Next.	Parallax illustrations, Wishy narrates in speech bubbles.
S03	Sign in / up	Phone, email, Google, Apple, guest.	Country picker spins like a wheel; fields wobble on error.
S04	OTP entry	6-digit code, resend timer, auto-read SMS (Android SMS Retriever).	Each digit pops into a bubble; success: bubbles fly up.
S05	Profile setup	Name, avatar, colour.	Avatar carousel with 3D tilt; colour choice floods the screen.
S06	Home (My Lists)	Pinned and other lists, search, Stardust and streak chip.	Staggered card entrance; cards float slightly (idle).
S07	List screen	Four views, filters, add wish, list menu.	Morph toggle between views.
S08	Vision Board	Free canvas editing, toolbar, stickers, export.	Objects lift and settle, washi tape stretches, auto-arrange spring.
S09	Wish detail	Image carousel, title, price, priority, note, savings meter, claim state (viewer), reactions.	Hero fly, parallax, flip to details.
S10	Add wish sheet	Camera, Gallery, Paste, Quick note.	Bubbles fly out from the Add button.
S11	Camera capture	In-app camera with grid, flash, switch lens.	Shutter flash and photo drops into a polaroid.
S12	Image editor	Crop, rotate, filters, sticker cut-out, frame preview.	Scissors animation traces the cut-out outline.
S13	Edit wish	Form fields with live card preview.	Card preview updates live and wobbles when changed.
S14	Savings screen	Entries, add amount keypad, Reveal Meter, estimated date.	Unblur reveal, coin rain.
S15	Grant flow	Date, price paid, memory photo, note.	Slide to grant: a slider pulls a star that ignites.
S16	Celebration	Full-screen celebration then share card.	Confetti cannon, stamp slam, screen shake.
S17	Wish Jar	Jar, shake, history of drawn wishes, settings.	Items float and clink inside the jar with physics.

ID	Screen	Purpose and key elements	Signature motion
S18	Granted Shelf	Trophy shelf, totals, filter by year.	Cards shine with periodic gloss sweep; shelf tilts with device (gyroscope parallax).
S19	Share list	Link, invite, role manager, revoke.	Paper plane flies out on send.
S20	Shared list (viewer)	Read-only board, claim button, reactions.	Claim button turns into a ribbon-wrapped gift.
S21	Me (profile)	Avatar, level, Stardust, streak, badges preview, stats.	Level ring fills; avatar waves.
S22	Trophy Room	All badges, locked and unlocked.	Locked badges wiggle; unlock plays a burst.
S23	Notifications inbox	Notifications list with swipe to dismiss.	Swipe sends items off like paper planes.
S24	Settings	Account, notifications, appearance (themes, fonts, icons), sound and haptics, data saver, privacy, help.	Shared-axis zoom, live preview of theme and font.
S25	Theme and font picker	Gallery with live preview card.	Circular reveal theme change.
S26	Bin	Deleted lists and wishes, restore, delete forever.	Paper balls rest in the bin; restore uncrumples them.
S27	Level-up and badge pop-ups	Overlay dialogs.	Burst, ribbon unfurl, sparkle rain.
S28	Wrapped (yearly)	Story-style recap.	Slide transitions, animated counters.
S29	Offline / error / maintenance	Messaging with retry.	Sad Wishy, wobbling plug.
S30	Admin console (web)	Dashboard, moderation, users, config.	Standard, no special motion.

8. Data Model

The following entities define the PostgreSQL schema. All tables use UUID version 7 primary keys (time-ordered), created_at and updated_at timestamps (UTC, timestamptz), and a nullable deleted_at for soft deletion. The client mirrors the same entities in Drift with additional sync columns: `sync_state` (synced, pending, conflict), `local_updated_at` and `server_version`.

Table	Main columns
users	id, display_name, handle (unique), email (nullable, unique), phone_e164 (nullable, unique), password_hash (nullable), avatar_type, avatar_ref, favourite_color, locale, timezone, currency_default, stardust_total, level, status (active, suspended, deleted), last_active_at
auth_identities	id, user_id, provider (phone, email, google, apple), provider_subject, verified_at
devices	id, user_id, platform (android, ios), push_token, app_version, os_version, perf_tier, last_seen_at
refresh_tokens	id, user_id, device_id, token_hash, expires_at, revoked_at, rotated_from
lists	id, owner_id, name, icon, cover_type, cover_value, description, privacy (private, invite, link), share_token_hash, target_date, sort_order, pinned, archived_at, board_settings (jsonb)
list_members	id, list_id, user_id, role (collaborator, viewer), invited_by, accepted_at, removed_at
wishes	id, list_id, creator_id, title, note, why_text, price_amount (numeric 12,2), price_currency, priority (1-5), category, mood, frame_style, product_url, status (open, granted, maybe_later), granted_at, granted_price, granted_note, granted_photo_id, savings_goal_amount, sort_order
wish_tags	wish_id, tag
images	id, owner_id, wish_id (nullable), storage_key, original_key, mime, width, height, bytes, blurhash, dominant_color, cutout_key (nullable), moderation_state (pending, ok, flagged, removed), position, status (uploading, processing, ready, failed)
board_items	id, list_id, wish_id (nullable), kind (wish, text, emoji, tape), content (jsonb), x, y, scale, rotation, z_index, version
savings_entries	id, wish_id, user_id, amount, currency, entry_date, note (append-only)
claims	id, wish_id, claimer_id, reveal_name (bool), released_at (hidden from Owner at API level)
reactions	id, wish_id, user_id, emoji, created_at
comments	id, wish_id, author_id, body, secret_to_viewers (bool), hidden_at
stardust_events	id, user_id, action, amount, ref_id, created_at
streaks	user_id, current_days, longest_days, last_day, freezes_left, freeze_reset_at
badges	id, code, name, description, rule (jsonb), icon_ref, rarity
user_badges	user_id, badge_id, unlocked_at
notifications	id, user_id, type, payload (jsonb), read_at, created_at
notification_prefs	user_id, category, enabled, quiet_start, quiet_end
reports	id, reporter_id, target_type, target_id, reason, status (open, actioned, dismissed), handled_by, handled_at

Table	Main columns
blocks	blocker_id, blocked_id, created_at
audit_log	id, actor_id, actor_role, action, target_type, target_id, ip, created_at, meta (jsonb)
feature_flags	key, value (jsonb), rollout_percent, updated_by
sync_tombstones	id, entity, entity_id, user_id, deleted_at (supports deletion sync to offline devices)

8.1 Key relationships

- A user owns many lists. A list has many wishes, many board items and many members.
- A wish has up to 5 images, many savings entries, many reactions and comments, and at most one active claim per claimer (many claimers allowed if the Owner enables Group Gift in a later release).
- A user has many devices, many notifications, many badges and one streak record.
- Claims are never returned by any endpoint callable by the list Owner. Database-level row security shall back this up (section 12).

8.2 Indexes and constraints

- Unique: users.handle, users.email, users.phone_e164, auth_identities(provider, provider_subject), list_members(list_id, user_id), user_badges(user_id, badge_id).
- Index: wishes(list_id, status, sort_order), wishes(creator_id, updated_at), images(wish_id, position), notifications(user_id, created_at desc), stardust_events(user_id, created_at).
- Full-text search: GIN index on a tsvector built from wishes.title, wishes.note and tags (use the simple configuration to support mixed languages).
- Check constraints: priority between 1 and 5, price_amount non-negative, rotation between -360 and 360, name length limits as in section 5.
- Foreign keys use ON DELETE CASCADE only for owned children (images, board_items, savings_entries). All other deletions go through soft delete and a scheduled hard-delete job.

9. API Specification

9.1 General conventions

Topic	Rule
Base URL	https://api.wishlistapp.example/api/v1 (production host to be confirmed by DevOps). Staging and development use their own sub-domains.
Format	JSON (UTF-8) request and response bodies. Dates and times in ISO 8601 UTC. Money as a string decimal with a separate currency code to avoid floating point errors.
Authentication	Authorization: Bearer <access token>. Access tokens last 15 minutes. Refresh tokens last 30 days, rotate on every use and are revoked on reuse (token theft detection).
Versioning	Version in the path (/v1). Breaking changes require /v2. The server shall expose the minimum supported app version in /config so old clients can be asked to update.
Pagination	Cursor-based: ?limit=30&cursor=<opaque>. Responses contain items, next_cursor and has_more.
Idempotency	All POST endpoints that create resources accept an Idempotency-Key header (UUID). Replays within 24 hours return the original result. The mobile sync queue shall always send this header.
Rate limiting	Per user and per IP, returned in headers X-RateLimit-Limit, X-RateLimit-Remaining, X-RateLimit-Reset. HTTP 429 with Retry-After when exceeded.
Compression	Gzip or Brotli on responses larger than 1 KB.
Caching	ETag and If-None-Match on read endpoints for lists and wishes; images served with immutable cache headers from the CDN.
Documentation	OpenAPI 3.1 generated by FastAPI, published at /docs on non-production environments, and exported as a build artefact for the frontend team to generate Dart client code.

9.2 Standard error format

Every error response shall use HTTP status codes correctly and this JSON body:

```
{
  "error": {
    "code": "WISH_NOT_FOUND",
    "message": "That wish does not exist or you cannot see it.",
    "details": { "wish_id": "0192f6c4-..." },
    "request_id": "req_8f1c2a"
  }
}
```

The complete list of error codes is in Appendix A. The frontend shall map each code to a friendly, localised message and an animation (for example the shaking field or the sad Wishy). Raw server messages shall never be displayed to users.

9.3 Endpoint catalogue

Authentication and account

Method	Path	Description	Auth
POST	/auth/otp/request	Send an SMS OTP to a phone number (E.164).	No
POST	/auth/otp/verify	Verify OTP; returns tokens and whether the user is new.	No
POST	/auth/register	Register with email and password.	No
POST	/auth/login	Login with email and password.	No
POST	/auth/oauth/{provider}	Exchange Google or Apple ID token for tokens.	No
POST	/auth/refresh	Rotate refresh token and get a new access token.	Refresh
POST	/auth/logout	Revoke the current refresh token (and device push token).	Yes
POST	/auth/password/forgot	Start password reset (email link or SMS OTP).	No
POST	/auth/password/reset	Complete password reset.	No
GET	/me	Get own profile, level, Stardust, streak and settings.	Yes
PATCH	/me	Update profile (name, avatar, colour, locale, currency).	Yes
POST	/me/identities	Link another sign-in method.	Yes
DELETE	/me/identities/{id}	Unlink a sign-in method.	Yes
POST	/me/export	Request a data export (asynchronous; link via notification).	Yes
DELETE	/me	Request account deletion.	Yes
POST	/me/devices	Register or update a device and its push token.	Yes

Lists and members

Method	Path	Description	Auth
GET	/lists	Lists the user owns or belongs to (filters: archived, pinned).	Yes
POST	/lists	Create a list.	Yes
GET	/lists/{id}	Get a list with summary counts and board settings.	Yes
PATCH	/lists/{id}	Update list fields, privacy, order, pin or archive.	Owner
DELETE	/lists/{id}	Move list to Bin.	Owner
POST	/lists/{id}/restore	Restore a list from the Bin.	Owner
POST	/lists/{id}/duplicate	Duplicate a list with its wishes.	Owner
POST	/lists/{id}/share-link	Create or regenerate the secret link.	Owner
DELETE	/lists/{id}/share-link	Revoke the link.	Owner
GET	/shared/{token}	Resolve a shared link to a read-only list view (public, rate limited).	Optional
POST	/shared/{token}/join	Join a shared list as Viewer.	Yes
GET	/lists/{id}/members	List members and roles.	Member
POST	/lists/{id}/members	Invite a member by phone, email or user id.	Owner
PATCH	/lists/{id}/members/{uid}	Change a member role.	Owner
DELETE	/lists/{id}/members/{uid}	Remove a member (or leave a list).	Owner / self

Wishes, images and board

Method	Path	Description	Auth
GET	/lists/{id}/wishes	Wishes in a list with filters, sort and cursor pagination.	Member
POST	/lists/{id}/wishes	Create a wish (accepts image ids already uploaded).	Collab
GET	/wishes/{id}	Get a wish with images, savings summary, reactions, claim state for the caller.	Member
PATCH	/wishes/{id}	Update a wish.	Collab
DELETE	/wishes/{id}	Move wish to Bin.	Collab
POST	/wishes/{id}/move	Move or copy to another list.	Collab
POST	/wishes/{id}/grant	Grant the wish with optional details.	Owner
POST	/wishes/{id}/ungrant	Undo grant within 24 hours.	Owner
POST	/wishes/bulk	Bulk move, delete, tag, grant (max 100 ids).	Collab
POST	/images/upload-url	Request a pre-signed upload URL and image id (content type and size declared).	Yes
POST	/images/{id}/complete	Confirm upload; enqueue processing.	Yes
GET	/images/{id}	Get processing status and URLs (original, display, thumbnail).	Member
POST	/images/{id}/cutout	Request background removal; result via status or push.	Collab
DELETE	/images/{id}	Delete an image.	Collab
GET	/lists/{id}/board	Get all board items.	Member
PUT	/lists/{id}/board	Replace board state (with base version for conflict detection).	Collab
PATCH	/lists/{id}/board/items/{item}	Update one board item (position, scale, rotation, z).	Collab
POST	/lists/{id}/board/export	Server-side render of the board to PNG (optional fallback).	Member

Savings, claims, reactions and the Jar

Method	Path	Description	Auth
GET	/wishes/{id}/savings	Entries, total, percentage, estimated completion date.	Owner
POST	/wishes/{id}/savings	Add a savings entry.	Owner
DELETE	/wishes/{id}/savings/{entry}	Remove a savings entry.	Owner
POST	/wishes/{id}/claim	Claim a wish (body: reveal_name).	Viewer
DELETE	/wishes/{id}/claim	Release own claim.	Viewer
PUT	/wishes/{id}/reaction	Set or change own emoji reaction.	Member
GET	/wishes/{id}/comments	List comments visible to the caller.	Member
POST	/wishes/{id}/comments	Add a comment (optionally secret to viewers).	Member
GET	/jar/draw	Draw a random open wish (weighted by priority) from selected lists.	Yes
GET	/jar/history	Previously drawn wishes.	Yes

Gamification, notifications, sync and misc

Method	Path	Description	Auth
GET	/gamification/summary	Level, Stardust, streak, freezes, weekly challenge.	Yes
POST	/gamification/streak/freeze	Use a Streak Freeze.	Yes
GET	/badges	Badge catalogue with the caller's unlock state.	Yes
GET	/notifications	Inbox (cursor pagination).	Yes
POST	/notifications/read	Mark notifications as read.	Yes
GET / PUT	/notifications/prefs	Read or update notification preferences and quiet hours.	Yes
GET	/sync/pull	Changes since a cursor for all entities the user can access, including tombstones.	Yes
POST	/sync/push	Batch of queued client mutations, each with an Idempotency-Key and client timestamp.	Yes
POST	/reports	Report a list, wish, comment or user.	Yes
POST	/blocks	Block a user.	Yes
GET	/config	Remote config, feature flags, minimum app version, theme and font manifest.	No
GET	/wrapped/{year}	Yearly summary data.	Yes
GET	/health	Liveness and readiness (used by orchestrator, no auth).	No

Admin API (separate prefix /admin/v1, staff tokens only, 2FA enforced)

Method	Path	Description
GET	/admin/v1/reports	Moderation queue with filters.
POST	/admin/v1/reports/{id}/action	Dismiss, hide, remove, warn, suspend.
GET	/admin/v1/users	Search users by handle, email, phone (masked).
POST	/admin/v1/users/{id}/suspend	Suspend or restore a user.
GET	/admin/v1/metrics	Dashboard aggregates.
GET / PUT	/admin/v1/flags	Read and update feature flags.
GET	/admin/v1/audit	Audit log search.
GET / POST / PUT	/admin/v1/badges	Manage badge catalogue.

9.4 Image upload flow

1. Client compresses the image (long edge at most 1600 px, JPEG or WebP quality about 80, EXIF location removed) and computes a checksum.
2. Client calls POST /images/upload-url with mime type, byte size and checksum. Server validates limits and returns an image id and a pre-signed PUT URL valid for 10 minutes.
3. Client uploads directly to object storage (resumable if supported). If offline, the upload stays in the queue.
4. Client calls POST /images/{id}/complete. Server verifies the object exists, marks status processing and enqueues the worker.
5. Worker validates the file signature (not just the extension), re-encodes to WebP, generates display (1080 px) and thumbnail (360 px) sizes, a BlurHash and a dominant colour, strips metadata, submits to automated moderation, then sets status ready.
6. Client learns of completion by WebSocket message, push or polling, and swaps the BlurHash placeholder for the real image with a fade and pop.

9.5 Real-time channel

A WebSocket endpoint `/ws/lists/{id}` authenticated with the access token provides live presence and board updates to members who have the list open. Messages are JSON with a type field: presence.join, presence.leave, cursor.move (throttled to 15 per second), board.item.update, wish.created, wish.updated, wish.granted and image.ready. The server shall drop connections idle for 60 seconds without a ping and the client shall reconnect with jittered exponential back-off. All state received over WebSocket is also obtainable by REST so the channel can fail without data loss.

9.6 Sample wish payload

```
GET /wishes/0192f6c4-7a11-7c3e-9d2b-1a5d3c0e9a10

{
  "id": "0192f6c4-7a11-7c3e-9d2b-1a5d3c0e9a10",
  "list_id": "0192f6a0-0f4a-7b1c-8e6d-5b3a4c2d1e0f",
  "title": "Gaming laptop",
  "note": "16GB RAM, at least 512GB SSD",
  "why_text": "For my final year project and for fun.",
  "price": { "amount": "7500.00", "currency": "GHS" },
  "priority": 5,
  "status": "open",
  "frame_style": "polaroid",
  "mood": "dream",
  "tags": ["tech", "school"],
  "images": [
    { "id": "0192f6c5-...", "position": 0, "status": "ready",
      "blurhash": "LEHV6nWB2yk8pyo0adR*.7kCMdnj",
      "urls": { "thumb": "https://cdn.../t.webp",
                "display": "https://cdn.../d.webp" } }
  ],
  "savings": { "total": "1875.00", "percent": 25.0,
               "estimated_completion": "2027-03-14" },
  "my_claim": null,
  "reactions": { "fire": 3, "heart": 2 },
  "created_at": "2026-11-05T09:14:22Z",
  "updated_at": "2026-11-05T09:14:22Z",
  "server_version": 12
}
```

10. Implementation Guidelines

10.1 Backend (Python)

Project structure

```
backend/
  app/
    main.py                 # FastAPI app factory, middleware, routers
    core/                   # config (pydantic-settings), security, logging
    db/                     # engine, session, base, migrations (alembic)
    modules/
      auth/                 # router.py, service.py, schemas.py, models.py
      users/
      lists/
      wishes/
      images/               # upload, pipeline tasks
      board/
      savings/
      claims/
      sharing/
      gamification/
      notifications/
      sync/
      admin/
    workers/                # celery app and tasks
    realtime/               # websocket manager (Redis pub/sub)
  tests/
  docker/
  pyproject.toml
```

ID	Requirement (the system shall...)	Pri.
BE-1	use a layered design per module: router (HTTP only), service (business rules), repository (database access). Routers shall not contain business logic and services shall not build HTTP responses.	M
BE-2	use asynchronous database access (SQLAlchemy async with asyncpg) and asynchronous HTTP clients. Blocking work such as image processing shall run in Celery workers, not in request handlers.	M
BE-3	validate every input with Pydantic models and reject unknown fields. Output models shall be explicit so that internal fields (password_hash, claims for Owners) cannot leak.	M
BE-4	enforce authorisation in a single reusable dependency (for example require_list_role(list_id, min_role)), and enforce claim secrecy at the query level: the Owner-facing wish query shall never join the claims table.	M
BE-5	manage the schema only with Alembic migrations. Migrations shall be backwards compatible with the previous application version to allow zero-downtime deployments (expand, migrate, contract pattern).	M
BE-6	implement the sync engine so that pull returns changes ordered by a monotonically increasing server_version per user scope, and push applies mutations idempotently and returns per-mutation results (applied, conflict with the server copy, rejected with code).	M

ID	Requirement (the system shall...)	Pri.
BE-7	run the image pipeline as chained Celery tasks (validate, re-encode, derive sizes, blurhash, moderate, finalise) with retries, dead-letter handling and a maximum processing time of 60 seconds per image.	M
BE-8	implement Stardust, streak and badge evaluation as event-driven jobs: domain events are written to the stardust_events table within the same transaction as the triggering change and evaluated asynchronously. Streak day boundaries shall use the user's stored time zone.	M
BE-9	implement rate limits with Redis (sliding window). Minimum limits: OTP request 3 per phone per 10 minutes and 10 per IP per hour; login 5 per minute per account; general authenticated traffic 120 requests per minute per user; share-link resolution 30 per minute per IP.	M
BE-10	send structured JSON logs including request_id, user_id (hashed where required), route, status and latency. No personal data, tokens or full phone numbers in logs.	M
BE-11	provide automated unit and integration tests with a minimum of 80 percent line coverage on services and 100 percent coverage on authorisation dependencies and claim secrecy.	M
BE-12	provide a seed script that creates realistic demo data (users, lists, wishes with sample images) for development and for app store review accounts.	S
BE-13	abstract third-party providers (SMS, push, moderation, object storage) behind interfaces so that vendors can be swapped by configuration only.	M

10.2 Frontend (Flutter)

Project structure (feature-first)

```
lib/
  main_dev.dart  main_staging.dart  main_prod.dart   # flavors
  app/                # app widget, router (go_router), theme wiring
  core/
    motion/           # motion_tokens.dart, transitions/, haptics.dart
    theme/            # colors, typography, font_packs, icon_packs
    widgets/          # sticker_button, squash_tap, wish_card, ...
    network/          # dio client, interceptors, api errors
    storage/          # drift database, secure storage, image cache
    sync/             # sync engine, queue, conflict rules
    audio/            # sound manager
    analytics/        # event tracker abstraction
  features/
    onboarding/  auth/  lists/  wishes/  board/  jar/
    shelf/  savings/  sharing/  profile/  gamification/
    notifications/  settings/
      data/ (api, dao, repository impl)
      domain/ (entities, repository interfaces, use cases)
      presentation/ (screens, widgets, controllers)
  l10n/               # ARB files
assets/ fonts/ lottie/ rive/ sounds/ images/
```

Concern	Package or approach
State management	flutter_riverpod with riverpod_generator; immutable state with freezed.
Navigation and deep links	go_router; app links for https share URLs and custom scheme wishlist://.
HTTP	dio, retrofit-style generated client from OpenAPI, interceptors for auth refresh and Idempotency-Key.
Local database	drift with a sync queue table; optional SQLCipher.
Secure storage	flutter_secure_storage for tokens.
Images	image_picker, camera, image_cropper or custom editor, flutter_image_compress, cached_network_image with disk cache, blurhash_dart.
Animation	flutter_animate, rive, lottie, confetti (or custom), simple_animations, custom painters, shader assets.
Sensors and haptics	sensors_plus (shake), vibration or native haptic channel, audioplayers or just_audio (low-latency pool).
Notifications	firebase_messaging, flutter_local_notifications.
Auth	google_sign_in, sign_in_with_apple, local_auth (biometrics), sms_autofill / SMS Retriever API.
Sharing	share_plus, screenshot or RepaintBoundary to image for board export.
Quality	very_good_analysis lints, mocktail, golden_toolkit, integration_test, patrol (optional), sentry_flutter.
Localisation	flutter_localizations and intl with ARB files; no hard-coded strings in widgets.

ID	Requirement (the system shall...)	Pri.
FE-1	follow feature-first clean architecture: presentation depends on domain, data implements domain interfaces, and widgets never call network or database classes directly.	M
FE-2	make the local database the single source of truth for the UI. Screens read from Drift streams; the sync engine updates the database; user actions write to the database and enqueue a mutation.	M
FE-3	build the Vision Board using a custom InteractiveViewer-style canvas with per-item gesture recognisers, hit testing in rotated coordinates and spring-based settling. Board state shall be serialised to the same JSON structure as board_items.	M
FE-4	build the Wish Jar using a lightweight 2D physics simulation (circle collision with gravity driven by device accelerometer) in a single CustomPainter or a Flame-like fixed-step loop capped at 30 active bodies on Medium and 12 on Low tier.	M
FE-5	compress and strip metadata from images on the device before upload, and generate a local thumbnail so the UI never waits for the server.	M

ID	Requirement (the system shall...)	Pri.
FE-6	create reusable widgets (SquashTap, StickerCard, PopIn, Stagger, WobbleOnPress, ConfettiBurst, RollingNumber, JellyTabBar) and use them everywhere so that motion stays consistent. Feature code shall not call AnimationController directly unless building a new reusable widget.	M
FE-7	provide three build flavors (dev, staging, prod) with separate application ids, icons, Firebase projects and API hosts.	M
FE-8	keep the Android download size under 35 MB and the installed size under 120 MB using app bundles, split ABIs, font subsetting, WebP assets and shrinking of Rive and Lottie files.	M
FE-9	ship with golden (screenshot) tests for key components in all themes and font packs, and integration tests covering the critical flows in section 14.	M
FE-10	log analytics events listed in Appendix B through an abstraction layer; events are sent only after the user has accepted analytics consent.	M
FE-11	provide a deterministic Demo mode (offline, fake data) for design reviews and store screenshots.	S

11. Non-Functional Requirements

11.1 Performance

ID	Metric	Target (reference device: Android, 3 GB RAM, mid-range)
NFR-P1	Cold start to interactive Home	At most 2.5 s (animated splash may continue but must not block input beyond this).
NFR-P2	Warm start	At most 1.0 s.
NFR-P3	Frame rate	60 fps sustained; below 1 percent janky frames in scripted scroll, drag and transition benchmarks.
NFR-P4	API latency (server side)	p95 below 300 ms for reads and below 500 ms for writes at 1,000 requests per second.
NFR-P5	Image upload	A compressed image (about 400 KB) shall upload in under 6 s on a 3G connection (about 1 Mbps).
NFR-P6	Image display	Thumbnails visible within 300 ms from disk cache and within 1.5 s from the network on 3G, with BlurHash placeholder shown immediately.
NFR-P7	Memory	Peak below 350 MB during Vision Board editing with 60 images; no leaks across repeated navigation.
NFR-P8	Battery	Less than 4 percent drain per 15 minutes of active use; zero background CPU when idle.
NFR-P9	Data usage	A typical 10-minute session on already-synced content shall use under 3 MB of mobile data.
NFR-P10	Sync	Incremental sync of 100 changed records shall complete in under 3 s on 4G.

11.2 Scalability, availability and durability

ID	Requirement
NFR-S1	Launch capacity: 20,000 registered users and 5,000 daily active users. Architecture shall scale horizontally to 500,000 registered users by adding API and worker instances and a PostgreSQL read replica, without code change.
NFR-S2	Availability target: 99.5 percent monthly at launch, rising to 99.9 percent. Planned maintenance shall be announced at least 24 hours ahead and shall not take the app offline, because the client works offline.
NFR-S3	Backups: automated daily full backups and continuous WAL archiving with point-in-time recovery for 7 days. Recovery point objective 15 minutes, recovery time objective 4 hours. Restores shall be tested quarterly.
NFR-S4	Object storage: versioning or replication enabled, lifecycle rules to purge hard-deleted images after 30 days.
NFR-S5	Stateless API containers with graceful shutdown, readiness and liveness probes, and at least two replicas in production.
NFR-S6	Graceful degradation: if the image worker, push provider or SMS provider is down, core list and wish features shall continue to work.

11.3 Usability and accessibility

- A new user shall be able to add their first wish within 60 seconds of opening the app, measured in usability tests with at least 5 participants, 80 percent of whom must succeed.
- Minimum touch target 48 by 48 dp. Gesture-only actions shall have a visible button alternative (important for accessibility).
- Contrast ratios and dynamic type as specified in sections 5.11 and 6.2. Every image shall allow an optional alt text, defaulting to the wish title.
- All animation shall be controllable by the Reduced Motion setting. No information shall be conveyed by motion or colour alone.

11.4 Compatibility and maintainability

- Supported OS versions as in section 2.3. Support the latest two major Flutter stable releases; upgrade within 6 weeks of each release.
- Support portrait orientation on phones in 1.0; tablet and landscape layouts are adaptive in 1.1.
- Code shall pass static analysis (zero warnings), formatting checks and dependency vulnerability scans in CI before merge.
- Every module shall have a README describing responsibilities, public interfaces and how to test it.

12. Security and Privacy Requirements

12.1 Application security

ID	Requirement (the system shall...)	Pri.
SEC-1	store passwords only as Argon2id hashes with per-user salts. Never log or return password hashes.	M
SEC-2	keep access tokens in memory and refresh tokens in the platform secure store (Keychain, Android Keystore). Revoke all tokens on password change and on account deletion.	M
SEC-3	require HTTPS everywhere. Android cleartext traffic shall be disabled; iOS App Transport Security shall remain enabled. Certificate pinning should be implemented for the API host with a documented rotation process.	M
SEC-4	protect against the OWASP API Security Top 10, in particular broken object-level authorisation. Every endpoint that takes a resource id shall verify the caller's relationship to that resource, and automated tests shall try accessing other users' resources for every such endpoint.	M
SEC-5	protect share links: tokens are random (at least 128 bits), stored only as hashes, rate-limited on resolution, revocable, and never included in analytics or logs.	M
SEC-6	keep image buckets private. Serve images through the CDN using short-lived signed URLs (at most 24 hours for display images). Share links shall not expose raw bucket URLs.	M
SEC-7	strip EXIF metadata (especially GPS location) from every uploaded image on the device and again on the server.	M
SEC-8	obfuscate release builds (Dart --obfuscate and --split-debug-info), and detect rooted or jailbroken devices as a signal only (no blocking), reducing risk to the claim secrecy feature by reporting to the backend for risk scoring.	S
SEC-9	store secrets in a secrets manager or encrypted environment variables, never in the repository. Rotate keys at least annually and immediately on suspected compromise.	M
SEC-10	complete an external penetration test and fix all critical and high findings before public launch.	M
SEC-11	scan dependencies (pip, pub) and container images in CI, and block merges with known critical vulnerabilities.	M
SEC-12	prevent OTP abuse (SMS pumping) through rate limits, per-country allow-lists, device attestation (Play Integrity, App Attest) on OTP requests, and spend alerts at the SMS provider.	M

12.2 Privacy and legal compliance

Wishlist collects personal data (phone number, email, name, photographs) and so must comply with the Ghana Data Protection Act, 2012 (Act 843) and, for users elsewhere, with the equivalent local law (for example GDPR for users in the European Economic Area and the UK). The product owner shall obtain legal advice to confirm final obligations before launch.

ID	Requirement (the system shall...)	Pri.
PRV-1	register the operating organisation as a data controller with the Data Protection Commission of Ghana before launch and renew the registration as required.	M
PRV-2	publish a plain-language Privacy Policy and Terms of Service, accessible before sign-up and from Settings, and record the version and time of the user's acceptance.	M
PRV-3	collect only the data needed for the features (data minimisation). Contacts, location and microphone permissions shall be requested only when the user triggers a feature that needs them, with a clear explanation.	M
PRV-4	provide users with access to, correction of, export of and deletion of their personal data, as specified in FR-A10 and FR-A11. Deletion requests shall be completed within 30 days; backups shall be purged on their normal rotation.	M
PRV-5	set a minimum age of 13 and, for users who state they are under 18, disable public link sharing by default and disable comments from non-members. An age gate shall appear at sign-up.	M
PRV-6	obtain consent for analytics and personalised notifications through an in-app consent screen. Declining shall not reduce core functionality.	M
PRV-7	maintain a register of all third-party processors (SMS, push, analytics, monitoring, moderation) with their data locations and legal agreements.	M
PRV-8	define and rehearse a data breach procedure: assess within 24 hours, notify the Data Protection Commission and affected users without undue delay as required by law.	M
PRV-9	complete the Google Play Data safety form and Apple App Privacy details accurately for every release.	M

13. Ghana Localisation and Market Requirements

Wishlist launches in Ghana first. The following requirements ensure the product feels native, works on real-world networks and devices, and is ready for the Ghanaian market.

Area	Requirement
Currency	Default currency GHS with the symbol GH₵ shown before the amount (for example GH₵ 7,500.00). Thousands separator comma, decimal point. Show pesewas only when non-zero in compact views.

Area	Requirement
Phone numbers	Accept local formats (024 123 4567, 054..., 055..., 059..., 020..., 050..., 026..., 027..., 057...) as well as +233 and normalise to E.164. The default country in the phone field is Ghana (+233).
Networks	OTP delivery shall be tested on MTN, Telecel and AirtelTigo SIMs. Fall back to a voice call or an email OTP if SMS is not delivered within 60 seconds (Should).
Data costs	Data Saver mode shall be offered on first launch (not buried in Settings), and the default image upload quality shall target 300 to 500 KB per image. Large Lottie and Rive assets shall download only on Wi-Fi.
Devices	QA device matrix shall include popular low and mid-range devices in Ghana (for example Tecno, Infinix, itel and Samsung Galaxy A-series phones with 2 to 4 GB RAM) and a minimum of two iPhones. The Low animation tier is expected to be the default on many of these devices and must still feel fun.
Power and connectivity	The app shall remain fully usable during power cuts and network outages (offline-first), and shall never lose a wish because of an interrupted upload.
Sharing	WhatsApp is the primary sharing channel. The share sheet shall place WhatsApp and Status first on Android where possible, and exported boards shall be sized for WhatsApp Status (9:16).
Mobile money	Plan the data model and the claim flow so that MoMo contributions (MTN MoMo, Telecel Cash, AirtelTigo Money) can be added in Release 1.2 through a licensed aggregator without redesign. Wishlist shall not hold customer funds.
Language	Release 1.0 is in English with Ghanaian-friendly microcopy and optional Pidgin-flavoured fun lines (Should). The localisation framework shall allow Twi, Ga, Ewe and Hausa packs in Release 1.1.
Cultural themes	Seasonal and cultural themes shall include Kente Pop (kente-inspired pattern background), Black Star (national colours), Independence Day (6 March), Harmattan Haze, Christmas, Easter and Eid. Cultural patterns shall be designed or licensed with respect and reviewed by Ghanaian reviewers.
Saving culture	Savings language shall feel familiar: for example the Susu Star badge and milestone phrases that reference susu saving habits, with no suggestion that Wishlist itself operates a susu or holds money.
Time and calendar	Ghana uses GMT (UTC+0) with no daylight saving. Streak and Daily Wish scheduling shall nevertheless use each user's stored time zone for diaspora users.
Support	In-app support through WhatsApp Business and email, with response within one business day.

14. Testing and Quality Assurance

14.1 Test levels

Level	Scope	Owner	Tooling
Unit	Services, repositories, view-models, motion helpers, sync rules.	Developers	pytest, flutter_test, mocktail
Widget and golden	Components in all themes, font packs, text scales and animation levels.	Frontend	flutter_test, golden_toolkit
Integration	API with real PostgreSQL, Redis and a storage emulator; app flows on emulator.	Both teams	pytest + testcontainers, integration_test
Contract	Mobile client against the OpenAPI spec, to catch breaking changes.	Both teams	schemathesis, generated client checks
End-to-end	Critical user journeys on real devices and in the cloud device farm.	QA	patrol or Maestro, Firebase Test Lab
Performance	Frame timing, startup, memory, API load and soak tests.	QA + DevOps	Flutter DevTools, k6 or Locust
Security	Authorisation tests, dependency scans, pen test, mobile static analysis.	Security / DevOps	Bandit, pip-audit, MobSF, external tester
Accessibility	Screen reader, large text, contrast and reduced motion.	QA + Design	TalkBack, VoiceOver, accessibility scanner
UAT / beta	Real users in Ghana (closed beta of at least 100 people).	Product owner	Google Play internal and closed tracks, TestFlight

14.2 Critical journeys (must pass before every release)

1. Install, onboard, sign up with a Ghana phone number OTP, and add the first wish within 60 seconds.
2. Add a wish with the camera while offline, close the app, go online, and confirm the wish and image sync correctly.
3. Switch between the four views and confirm that Vision Board positions persist and match across two devices.
4. Shake the Wish Jar and obtain a wish; grant it and see the celebration and the Shelf update.
5. Set a savings goal, add entries, and confirm the Reveal Meter reaches 25, 50, 75 and 100 percent with milestone pop-ups.
6. Share a list by link, open it on a second account, claim a wish, and confirm the Owner cannot see the claim in the app, the API, notifications or exports.

7 Revoke a share link and confirm the old link stops working immediately.

8 Change theme, font pack and animation level, and confirm live updates and persistence.

9 Delete a wish, undo it, delete again and restore from the Bin.

10 Request account deletion and confirm data removal and sign-out on all devices.

14.3 Animation and delight QA checklist

- Every transition in section 6.6 plays correctly forward and in reverse, and can be interrupted by a back gesture.
- No dropped frames in scripted runs on the reference device and on the lowest-tier device in the matrix.
- Reduced and Off animation levels remove decorative motion but keep every function usable.
- No flashing more than 3 times per second; verified with a photosensitive epilepsy analysis tool on celebration animations.
- Sounds play at balanced volume, respect silent mode and never delay the UI.
- No memory growth after 50 open-close cycles of any screen; no controller leaks reported by DevTools.

14.4 Sample acceptance criteria

The format below shall be used for all user stories (Given, When, Then). Examples:

```
FR-G3  Reveal Meter
Given a wish priced GH₵ 1,000.00 with GH₵ 250.00 saved (25 percent)
When I add a savings entry of GH₵ 250.00
Then the total shows GH₵ 500.00 (50 percent)
 And the image blur radius animates from its 25 percent value to its 50
     percent value within 900 ms
 And a 50 percent milestone pop-up appears and awards 50 Stardust.

FR-H4  Secret claim
Given Kwame owns the list "Birthday" and Ama is a Viewer
When Ama claims the wish "Sneakers"
Then Ama sees "You are getting this"
 And other viewers see "Claimed"
 And Kwame sees no claim indicator on any screen, API response, push
     notification, export or comment feed.

FR-L1  Offline add
Given the device is in airplane mode
When I add a wish with two photos
Then the wish appears immediately with sync state "pending"
 And when the device reconnects the wish and images upload and the state
     becomes "synced" without my intervention.
```

15. DevOps, Release and Operations

Topic	Requirement
Source control	Git with trunk-based development: short-lived feature branches, pull requests with at least one review, conventional commit messages, protected main branch. Separate repositories (or monorepo folders) for mobile, backend and admin.
CI (every pull request)	Backend: lint (ruff), type check (mypy), tests, migration check, dependency scan, container build. Mobile: format, analyze, unit, widget and golden tests, build debug APK. Both must pass before merge.
CD	Merge to main deploys to Development automatically. A release tag deploys to Staging; manual approval deploys to Production. Mobile builds uploaded by Fastlane to Play internal track and TestFlight.
Infrastructure as code	Terraform (or equivalent) for cloud resources; Docker images for all services; Docker Compose for local.
Observability	Sentry for backend and mobile crashes; Prometheus and Grafana for metrics; uptime checks on /health; alerts for error rate, latency, queue depth, failed uploads and SMS delivery rate.
Release management	Semantic versioning. Staged rollout on Google Play (5, 20, 50, 100 percent) with automatic halt on crash-free rate below 99 percent.
Forced update	The /config endpoint shall expose min_supported_version. Older app versions shall show a friendly update screen.
Incident response	On-call rota, severity levels, runbooks for common failures, and a blameless post-mortem within 5 working days of any severity 1 incident.
Cost control	Budgets and alerts on cloud, SMS and moderation spend. Image lifecycle rules and CDN caching to minimise egress.

16. Delivery Plan and Milestones

The following plan assumes a team of 2 Flutter developers, 2 Python developers, 1 UI and motion designer, 1 QA engineer and a part-time DevOps engineer. Durations are estimates and shall be re-baselined by the team after Phase 0.

Phase	Weeks	Backend deliverables	Frontend and design deliverables	Exit criteria
0. Foundations	1-2	Repository, CI/CD, Docker, PostgreSQL schema v1, auth skeleton, OpenAPI contract draft.	Flutter project and flavors, design system (colours, fonts, icons v1), motion token library, SquashTap and StickerCard widgets, clickable prototype.	Contract approved. Hello-world deployed to Development. Design system signed off.

Phase	Weeks	Backend deliverables	Frontend and design deliverables	Exit criteria
1. Core	3-6	Auth (OTP, email, social), lists, wishes, images pipeline, sync endpoints.	Onboarding, auth screens, Home, List (Grid), Add flow with camera and gallery, image editor basics, offline database and sync engine.	Critical journeys 1, 2 and 9 pass. Internal alpha on Play.
2. Delight	7-10	Savings, grant, jar draw, Stardust, streaks, badge engine.	Vision Board, Stack and Timeline views, Wish Jar physics, Granted celebration and Shelf, Reveal Meter, sticker cut-out, themes, font and icon packs, sound and haptics.	All animation tiers work. 60 fps verified. Journeys 3, 4, 5 and 8 pass.
3. Social	11-13	Sharing, members, claims with secrecy tests, reactions, comments, WebSocket, notifications, reports.	Share flows, shared list view, claim UI, collaboration cursors, notifications inbox, deep links.	Journeys 6 and 7 pass. Claim secrecy test suite green.
4. Hardening	14-16	Admin console, moderation, rate limits, backups tested, load test, penetration test fixes.	Accessibility pass, performance tuning on low-end devices, localisation framework, store assets, Data Saver.	Closed beta (100+ users in Ghana) with crash-free rate above 99 percent. Pen test criticals closed.
5. Launch	17	Production go-live, monitoring, support runbooks.	Play Store and App Store release with staged rollout.	Release criteria in section 18 met.

16.1 Release roadmap after 1.0

Release	Highlights
1.1	Product link import (title, price, image), Twi / Ga / Ewe / Hausa language packs, tablet layouts, exchange rates, Wrapped refinements.
1.2	Mobile money contributions from friends through a licensed aggregator, Group Gifts, voice notes, weekly challenges, Round-up Jar.
1.3	Web client for viewing and editing boards, AR preview of items, widget for the home screen (Daily Wish).

17. Risks and Mitigations

Risk	Likelihood	Impact	Mitigation
Heavy animation causes jank or overheating on low-end devices	High	High	Animation tiers (UI-A1), particle caps, early testing on Tecno, Infinix and itel phones, performance budgets as CI gates.
Large app size and high data usage discourage installs	Medium	High	Size budget (FE-8), downloadable asset packs on Wi-Fi, Data Saver on first launch.
SMS OTP delivery failures or SMS pumping fraud	Medium	High	Provider redundancy, voice and email fallback, rate limits, attestation, spend alerts (SEC-12).
Claim secrecy leak (Owner discovers claims)	Low	Very high	Query-level exclusion, dedicated test suite, code-owner review on any change touching claims, pen test scope.
Inappropriate images uploaded to shared lists	Medium	High	Automated moderation, reporting, blocking, quick takedown SLA, link sharing off by default for minors.
Offline sync conflicts cause data loss	Medium	High	Idempotent mutations, tombstones, per-field merge rules, conflict logging, extensive sync tests with simulated network faults.
Scope creep from the many fun features	High	Medium	Priority tags (M, S, C); phase gates; change control through the product owner.
Font, icon, sound or animation licensing problems	Medium	Medium	Use only OFL or properly licensed assets; asset register with licence and source for every item.
Regulatory non-compliance (data protection)	Low	High	Early registration with the Data Protection Commission, legal review, privacy by design (section 12.2).
Third-party service outage (push, SMS, storage)	Medium	Medium	Provider abstraction (BE-13), graceful degradation (NFR-S6), status monitoring.

18. Acceptance and Sign-off

18.1 Definition of Done (for each requirement)

- Implemented according to this SRD and reviewed through a pull request.
- Automated tests written and passing, with coverage targets met.
- Accessibility and animation checks completed (sections 11.3 and 14.3).
- Works offline where applicable, and tested on at least one low-end Android device.
- Analytics events added, error states designed, strings localised.
- Documentation updated (README, OpenAPI).

18.2 Release criteria for version 1.0

- All Must requirements implemented and verified; no open critical or high defects.
- All 10 critical journeys pass on the device matrix.
- Performance targets in section 11.1 met on the reference device.
- Crash-free sessions of at least 99.5 percent during the closed beta.
- Penetration test completed and critical and high findings fixed.
- Data Protection Commission registration and legal documents in place.
- Backups and restore procedure tested; monitoring and alerts live.
- App store listings, screenshots, privacy forms and review accounts ready.

18.3 Sign-off

Role	Name	Signature	Date
Product owner			
Lead Flutter developer			
Lead Python developer			
UI and motion designer			
QA lead			
DevOps engineer			

Appendix A: Error Codes

Code	HTTP	Meaning	Suggested UI behaviour
VALIDATION_FAILED	422	One or more fields invalid (details list the fields).	Shake the offending fields.
UNAUTHENTICATED	401	Missing or expired token.	Silent refresh, else sign-in screen.
FORBIDDEN	403	Role does not allow the action.	Friendly pop-up.
NOT_FOUND	404	Resource missing or not visible.	Sad Wishy page.
WISH_NOT_FOUND / LIST_NOT_FOUND	404	Specific resource missing.	Return to list.
CONFLICT	409	Edit conflict; server copy returned.	Merge silently or show conflict sheet.
OTP_INVALID	400	Wrong or expired code.	Digits wobble and clear.
OTP_RATE_LIMITED	429	Too many OTP requests.	Countdown timer.
RATE_LIMITED	429	Too many requests.	Back off; do not show an error unless persistent.
IMAGE_TOO_LARGE	413	Over the size limit.	Offer to compress.
IMAGE_UNSUPPORTED	415	Type not allowed.	Explain supported types.
IMAGE_REJECTED	422	Moderation rejected the image.	Explain policy; remove image.
LIMIT_REACHED	403	Soft limits (lists, wishes, images).	Friendly limit message.
LINK_REVOKED	410	Share link no longer valid.	Explain and ask the owner for a new link.
ACCOUNT_SUSPENDED	403	Account suspended.	Show appeal information.
UPDATE_REQUIRED	426	App version below minimum.	Update screen.
SERVER_ERROR	500	Unexpected error.	Retry and report with request_id.
SERVICE_UNAVAILABLE	503	Dependency down or maintenance.	Offline-friendly message.

Appendix B: Analytics Events

Events are sent only after consent (PRV-6), contain no free text typed by users, and use hashed user ids.

Event	Properties
app_open	cold_or_warm, perf_tier, animation_level
onboarding_step / onboarding_skip	step_index
signup_start / signup_complete	method (phone, email, google, apple, guest)
wish_add_start / wish_add_complete	source (camera, gallery, paste, share), image_count, duration_ms
view_changed	list_view (grid, stack, board, timeline)
board_edit	items_count, duration_s
jar_shake / jar_reveal / jar_action	action (open, put_back, grant)
savings_add / milestone_reached	percent_bucket
wish_granted	days_since_created, had_savings (bool)
list_shared / claim_made / claim_released	role, channel (whatsapp, copy, other)
badge_unlocked / level_up	badge_code, level
streak_extended / streak_lost / freeze_used	streak_days
settings_changed	setting_key, new_value (non-personal values only)
sync_result	duration_ms, items, outcome
perf_sample	p50_frame_ms, p95_frame_ms, screen, tier
error_shown	error_code, screen

Appendix C: Badge Catalogue

Code	Name	Unlock rule
first_spark	First Spark	Add your first wish.
akwaaba	Akwaaba	Complete onboarding and create your profile.
snap_happy	Snap Happy	Add 10 wishes with camera photos.
gallery_goblin	Gallery Goblin	Add 25 images from the gallery.
collector	Collector	Have 25 wishes in total.
dream_hoarder	Hoarder of Dreams	Have 100 wishes in total.
board_boss	Board Boss	Place 10 items on a Vision Board.
sticker_surgeon	Sticker Surgeon	Use the Sticker Cut-out tool 5 times.
frame_fanatic	Frame Fanatic	Use all 10 frame styles.
kente_collector	Kente Collector	Try 5 different themes.
shake_it	Shake It	Use the Wish Jar for the first time.
jar_junkie	Jar Junkie	Use the Wish Jar 30 times.
maybe_later	Maybe Later	Send 10 wishes to Maybe Later.
decisive	Decisive	Grant 5 wishes drawn from the Jar.
dream_come_true	Dream Come True	Grant your first wish.
hat_trick	Hat Trick	Grant 3 wishes in one calendar month.
ten_down	Ten Down	Grant 10 wishes.
legend	Legend	Grant 50 wishes.
penny_starter	Penny Starter	Log your first savings entry.
susu_star	Susu Star	Log savings in 12 consecutive weeks.
quarter_way	Quarter Way	Reach 25 percent on any wish.
halfway_hero	Halfway Hero	Reach 50 percent on any wish.
almost_there	Almost There	Reach 90 percent on any wish.
black_star_saver	Black Star Saver	Fully save for 5 wishes.
crystal_clear	Crystal Clear	Complete a Reveal Meter to 100 percent.
streak_3	Warm-up	3-day streak.
streak_7	On Fire	7-day streak.

Code	Name	Unlock rule
streak_30	Blazing	30-day streak.
streak_100	Rainbow Flame	100-day streak.
freeze_master	Freeze Master	Use a Streak Freeze.
sharer	Sharer	Share a list for the first time.
social_butterfly	Social Butterfly	Invite 5 people.
gift_guardian	Gift Guardian	Claim a wish that is later granted.
secret_santa	Secret Santa	Have 3 claims granted.
cheerleader	Cheerleader	Leave 20 reactions.
co_creator	Co-creator	Edit a shared Vision Board with another person.
night_owl	Night Owl	Add a wish between 00:00 and 04:00.
early_bird	Early Bird	Open the app before 07:00 on 7 different days.
harmattan_dreamer	Harmattan Dreamer	Add 5 wishes in December.
independence_star	Independence Star	Open the app on 6 March.
level_10	Level 10	Reach level 10.

Appendix D: Level Table (Initial Values)

Level	Title	Stardust needed (cumulative)	Unlock
1	Daydreamer	0	Default frames
3	Wisher	250	Frame: torn paper
5	Collector	800	Font pack: Handwritten
8	Dream Chaser	2,000	Theme: Neon Night
10	Star Catcher	3,200	Icon pack: Neon
15	Wish Wizard	7,500	Alternate app icon: Wizard
20	Dream Architect	14,000	Frame: gold leaf
25	Constellation Maker	24,000	Theme: Kente Pop Deluxe
30	Wish Legend	40,000	Animated avatar crown

The Stardust curve is configurable through remote config (FR-M6). Intermediate levels follow a smooth curve defined by the backend and returned by GET /gamification/summary

so the client never hard-codes thresholds.

Appendix E: Glossary of Fun

Term	Meaning
Wishy	The friendly floating star mascot who guides users and appears in empty states and errors.
Sticker Pop	The visual style of the app (thick outlines, bright colours, hard shadows).
Jelly	Motion style with squash, stretch and wobble.
Tier	Animation quality level (High, Medium, Low) chosen by device capability.
Tombstone	A record that marks something as deleted so that offline devices can remove it when they sync.
BlurHash	A compact string that represents a blurred preview of an image, used as an instant placeholder.

End of document SRD-WISH-001, version 1.0.